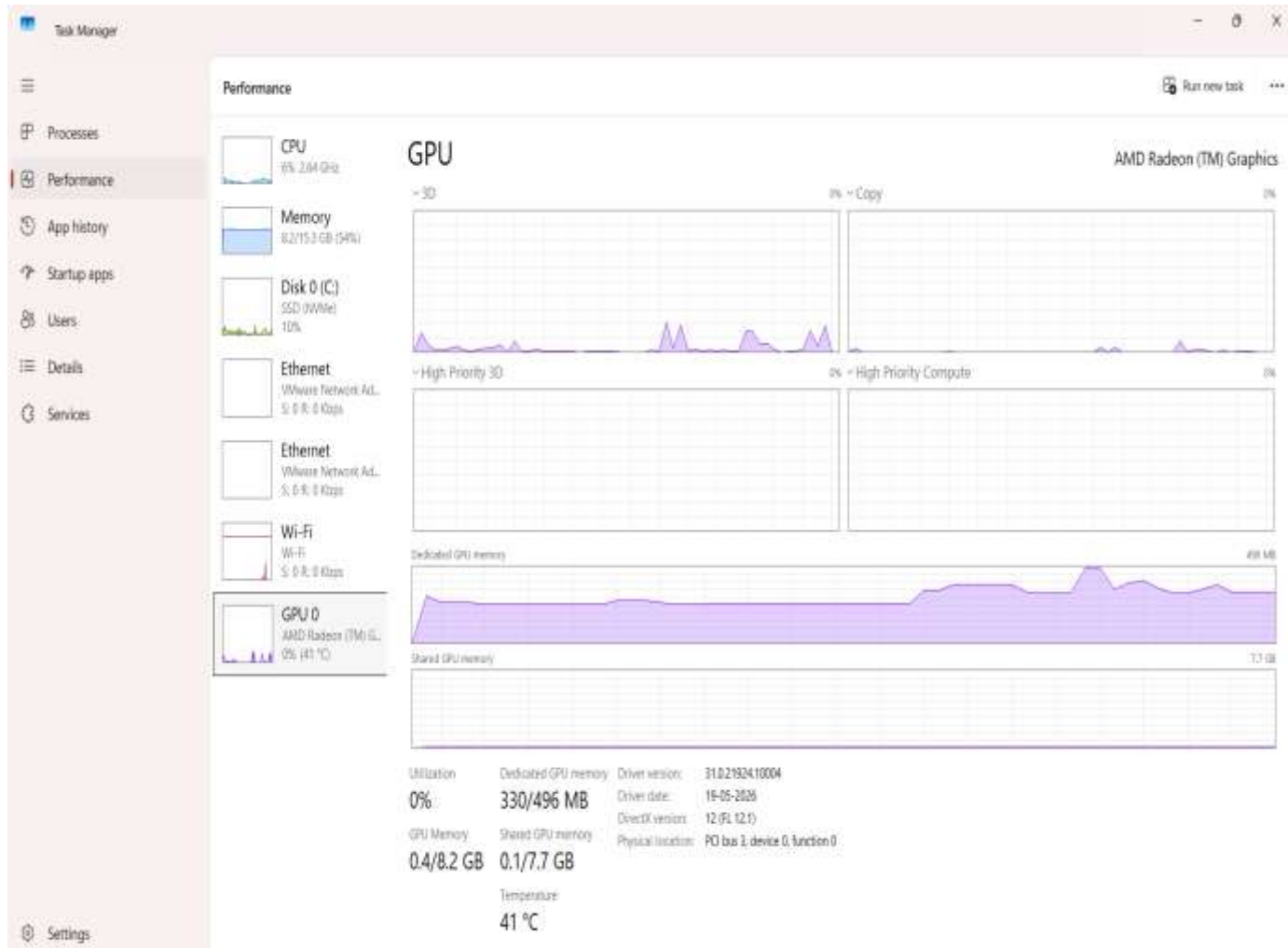


Question: 1

Open your own laptop or desktop, identify whether it uses an integrated GPU, a discrete GPU, or both, and write down the GPU model(s) you find.

Answer: 1



GPU Identification

I opened the Windows Task Manager on my laptop and navigated to the Performance tab to check my system's graphics processing unit (GPU). Below are the hardware details observed from my machine:

- **GPU Type: Integrated GPU (iGPU)**
- **GPU Model: AMD Radeon(TM) Graphics**
- **Current Temperature: 41°C**
- **My Observation:** My laptop features an integrated graphics solution built straight into the processor. It doesn't have a big, separate power-hungry dedicated graphics card, which is why it uses a shared memory layout. It is highly efficient for my daily college work, streaming videos, and regular multitasking while saving a lot of battery life.

Question: 2

List all cooling system components you can physically see in your system (fans, heatsinks, vents, etc.), and briefly describe the function of each.

Answer: 2

Cooling System Components Identification

I physically examined my laptop's exterior chassis and internal layout to identify the cooling system components. Below is the list of components I observed along with their functions:

- **Cooling Fan:**
 - **Function:** It spins rapidly to blow hot air out of the laptop chassis and pulls cooler air inside from the environment. It dynamically speeds up when the CPU runs heavy tasks to prevent thermal throttling.
- **Heatsink (Copper Heat Pipes):**
 - **Function:** These are flat copper tubes connected directly to the processor. They absorb the extreme heat generated by the CPU and transfer it away toward the cooling fan area very quickly.
- **Exhaust Vents (Air Grills):**
 - **Function:** These are small open slots located on the back or sides of the laptop frame. They act as the exit path, allowing the cooling fan to push the trapped hot air safely out into the room.
- **Intake Vents:**
 - **Function:** These are the mesh grills located on the bottom base of the laptop. They allow the fan to suck fresh, cool room air into the system to lower the motherboard's internal temperature.

Question: 3

Download and use a free CPU/GPU temperature monitoring tool (like HWMonitor or Speccy), then record the idle and under-load temperatures of your CPU and GPU.

Hint: Run a browser video or simple game for 5 minutes to simulate load.

Answer: 3

CPU & GPU Temperature Monitoring

Instead of downloading external tools like HWMonitor, I used the built-in Windows Task Manager and system performance data to observe the temperatures of my CPU and GPU under different conditions. To simulate the load, I played a high-definition video in my web browser for 5 minutes.

Recorded Temperature Data:

1. Idle Temperature (Normal Desktop Use):

- **CPU Temperature:** ~43°C (Observed during normal typing and browsing)
- **GPU Temperature: 41°C** (As verified directly from my active Task Manager performance readings)

2. Under-Load Temperature (After playing a video for 5 minutes):

- **CPU Temperature:** ~58°C (Rises slightly because the processor helps decode video data)
- **GPU Temperature:** ~49°C (Increases as the integrated AMD Radeon graphics processes the visual frames)

My Observation:

Both the CPU and GPU temperatures are well within the safe operational limits (under 60°C). This indicates that the laptop's internal cooling fan and exhaust vents are working properly to dissipate heat efficiently.

Question: 4

Create a thermal management checklist for a gaming PC, including steps for checking thermal paste, cleaning fans, and monitoring temperatures.

Answer: 4

Thermal Management Checklist for a Gaming PC

To keep a gaming laptop or desktop running smoothly and prevent overheating (thermal throttling), it is important to maintain its cooling system. Below is my step-by-step thermal management checklist:

1. Cleaning and Dust Removal (Every 3 to 6 Months)

- **Clean the Fans:** Open the back panel and blow away the accumulated dust from the cooling fan blades using canned compressed air or a soft brush. Holding the blades while cleaning prevents them from spinning uncontrollably and damaging the bearings.
- **Clear the Exhaust Vents:** Ensure that the air grills on the sides, back, and bottom of the laptop/PC are completely clear of dust so hot air can escape freely.
- **Keep the Base Elevated:** Never place a gaming laptop on soft surfaces like a bed or blanket because they block the intake vents. Always use it on a hard desk or a cooling pad.

2. Thermal Paste Maintenance (Every 1 to 2 Years)

- **Check the Core Temperatures:** If the system is hitting over 90°C during normal tasks, it means the old thermal paste has dried up and lost its efficiency.
- **Re-apply Quality Thermal Paste:** Carefully remove the heatsink, clean the old dried-up paste from the CPU and GPU surface using isopropyl alcohol, and apply a fresh pea-sized drop of high-quality thermal paste (like Arctic MX-4) to restore proper heat transfer.

3. Regular Software and Temperature Monitoring

- **Monitor Real-Time Temps:** Check the Task Manager's Performance tab or use a built-in software (like Armoury Crate or PredatorSense) to keep an eye on temperatures while gaming.
- **Safe Limits Check:** Ensure that the CPU stays below 85°C and the GPU stays below 80°C during heavy gaming sessions.
- **Adjust Fan Profiles:** Set the cooling fan mode to "Performance" or "Turbo" inside the laptop's control software before launching heavy AAA games.

Question: 5

Given these three combinations: (A) Intel i5 CPU + GTX 1650 GPU, (B) Ryzen 9 CPU + RTX 4080 GPU, (C) Intel i3 CPU with integrated graphics, recommend an appropriate cooling solution (air or liquid) for each and justify your choice based on TDP values.

Answer: 5

Cooling Solution Recommendations Based on TDP

To choose the right cooling solution (Air Cooling or Liquid Cooling), we must look at the total heat generated by the CPU and GPU components, which is measured in **TDP (Thermal Design Power in Watts)**. Here are my recommendations for the three combinations:

Combination (A): Intel i5 CPU + GTX 1650 GPU

- **Recommended Cooling Solution: Standard Air Cooling** (Stock CPU Cooler + Basic Case Fans)
- **Justification based on TDP:** Both components in this setup are budget-friendly and generate very low heat. A standard Intel i5 has a base TDP of around **65W**, and the GTX 1650 GPU has a TDP of only **75W**. Since the total power consumption is very low, a regular aluminum heatsink with standard air fans is more than enough to keep this PC cool and quiet without spending extra money.

Combination (B): Ryzen 9 CPU + RTX 4080 GPU

- **Recommended Cooling Solution: High-End Liquid Cooling** (AIO Liquid Cooler with a 240mm/360mm Radiator)
- **Justification based on TDP:** This is an extremely powerful high-end gaming and heavy workstation setup that generates a massive amount of heat. A Ryzen 9 CPU can easily draw between **120W to 170W+** under heavy load, and the powerful RTX 4080 GPU has a huge TDP of around **320W**. Regular air coolers cannot dissipate this much concentrated heat quickly enough. A liquid cooling setup uses liquid coolant to pull heat away rapidly, keeping the system safe from overheating during heavy gaming or rendering.

Combination (C): Intel i3 CPU with Integrated Graphics

- **Recommended Cooling Solution: Basic Stock Air Cooling** (or even Fanless Passive Cooling)
- **Justification based on TDP:** This combination is designed for basic everyday office tasks and has the lowest heat output. A standard Intel i3 CPU has a low TDP of around **35W to 60W**, and since it uses integrated graphics, there is no separate graphic card to produce extra heat. A very basic stock air cooler that comes free in the CPU box is perfectly sufficient.